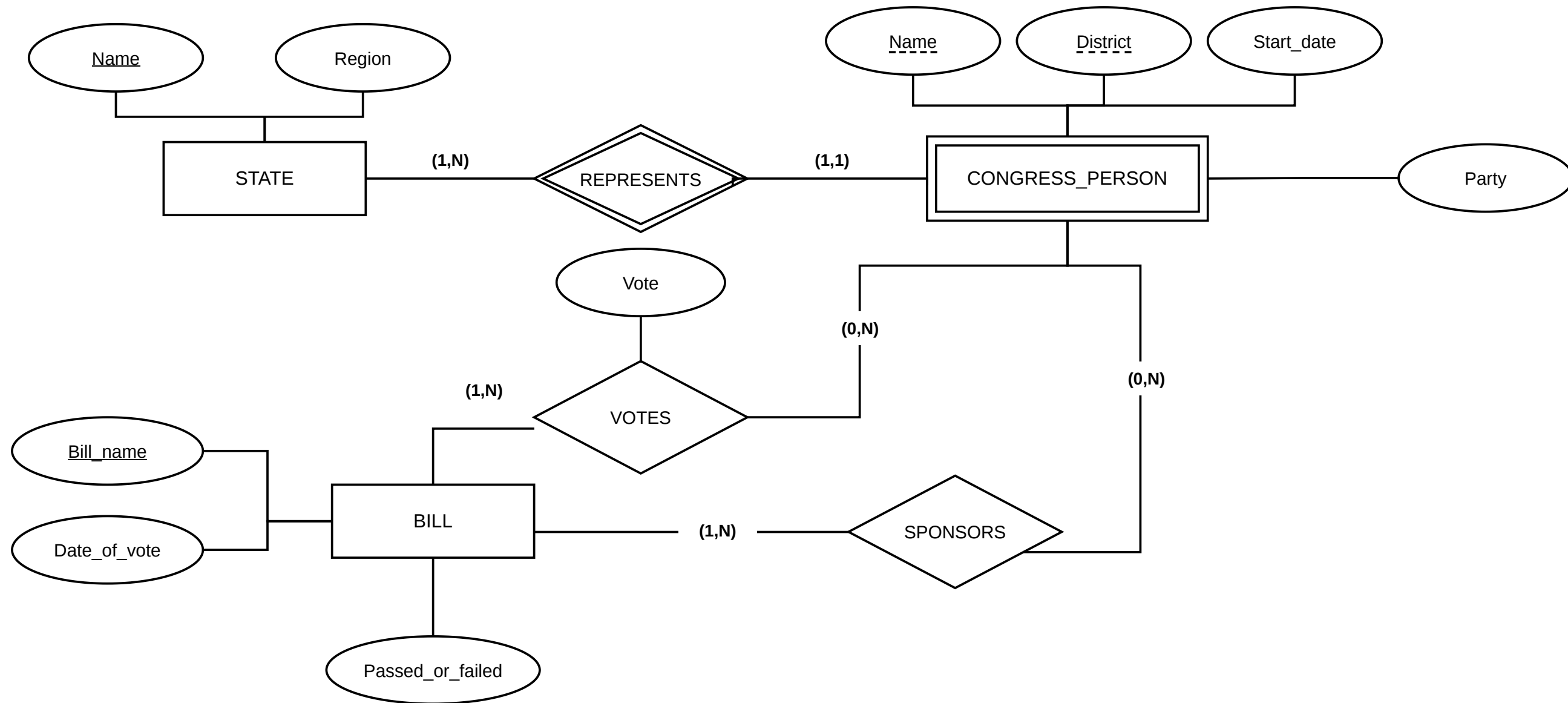


Solution 1

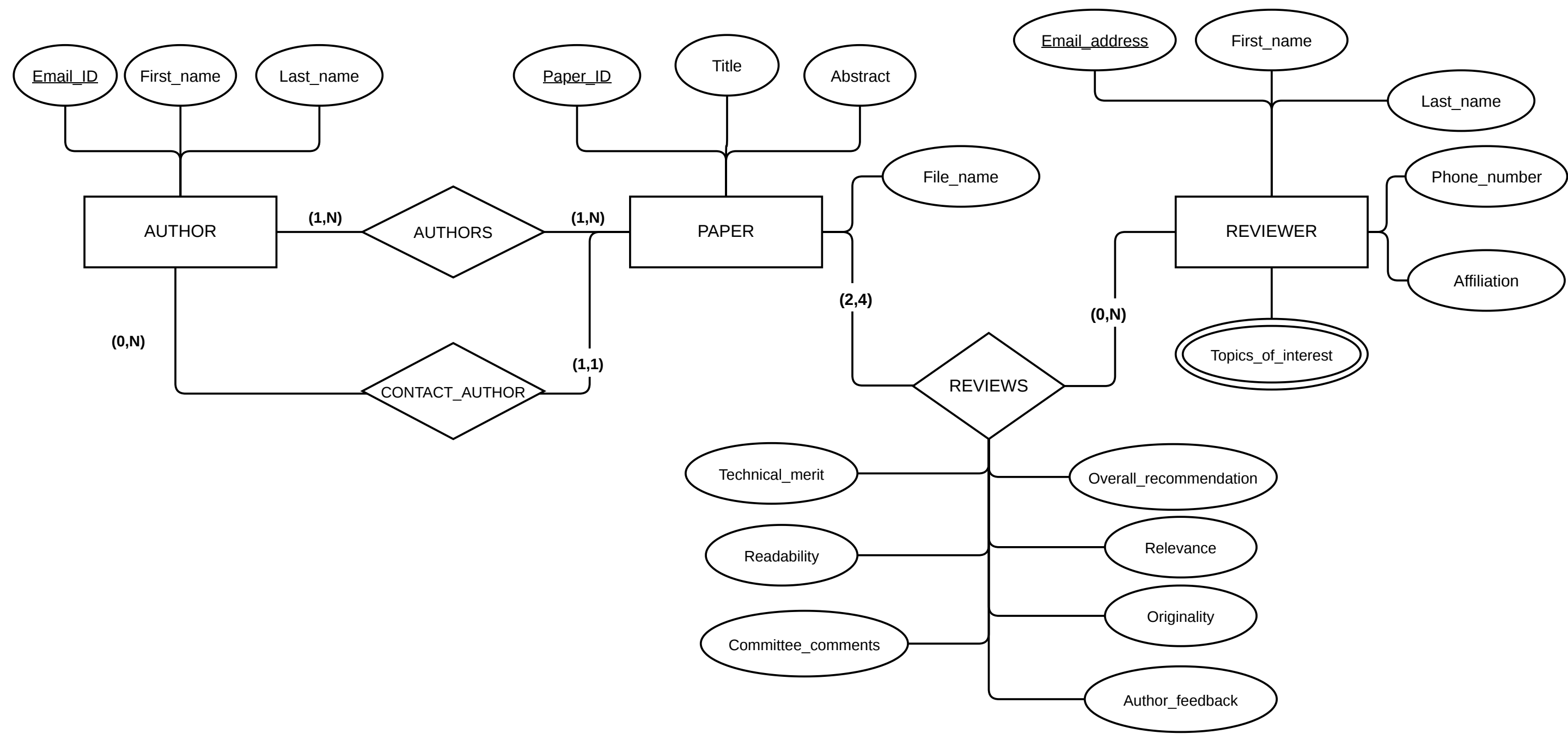


Assumptions and Domain Constraints

1. Each STATE has at least one CONGRESS_PERSON; each CONGRESS_PERSON represents exactly one STATE.
2. CONGRESS_PERSON is weak. Within a STATE, (Name, District) is its composite partial key; full key = (State.Name, Name, District).
3. Bill_name uniquely identifies a BILL in this database.
4. Every BILL has at least one sponsor; a CONGRESS_PERSON may sponsor zero or many bills.
5. Each eligible CONGRESS_PERSON has exactly one VOTES record per BILL considered during that person's service. "Absent" is stored as a vote value.
6. Only voting representatives of the 50 states are modeled; non-voting delegates are excluded.

Domains: Region is in {Northeast, Midwest, Southeast, Southwest, West}; Party is in {Republican, Democrat, Independent, Other}; Passed_or_failed is in {Yes, No}; Vote is in {Yes, No, Abstain, Absent}.

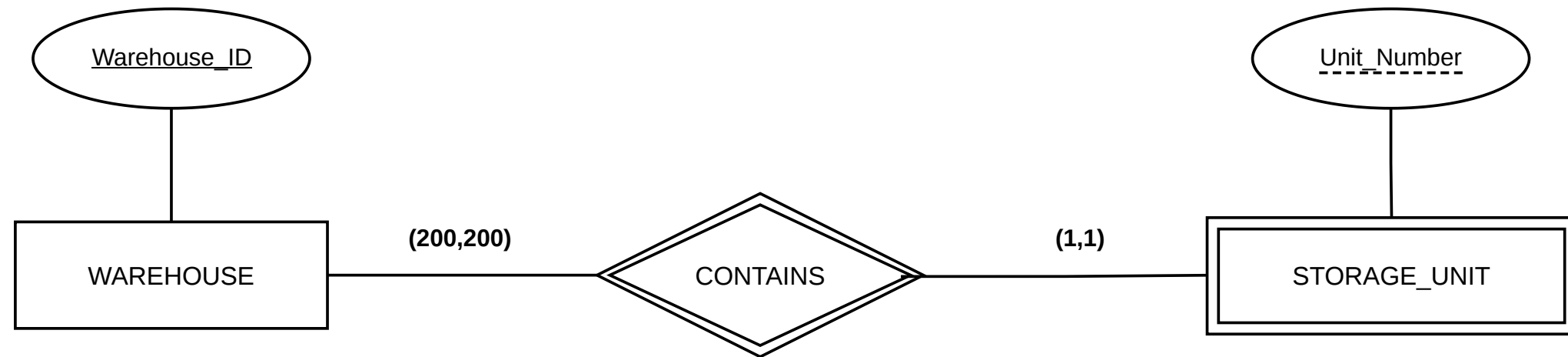
Solution 2



Assumptions and Domain Constraints

- 1. Every PAPER has at least one AUTHOR; authors are stored only when associated with a submitted paper.
- 2. Every PAPER has exactly one contact author, and that person must also participate in AUTHORS for the same PAPER.
- 3. Every PAPER is assigned to 2–4 distinct REVIEWERS. A REVIEWER may have no current assignment.
- 4. A REVIEWER reviews a particular PAPER at most once; therefore, (Paper_ID, Reviewer.Email_address) uniquely identifies a review occurrence.
- 5. Technical_merit, Readability, Originality, and Relevance are required integer ratings in {1,...,10}.
- 6. Overall_recommendation is assumed to have domain {Accept, Reject}. Committee_comments and Author_feedback are separate text values.
- 7. Topics_of_interest is multivalued. A person may be both an AUTHOR and a REVIEWER; the two roles are maintained separately.
- 8. Conflict-of-interest rules and final paper-selection status are outside the stated requirements.
- 9. Constraint: CONTACT_AUTHOR is subset of AUTHORS

Solution 3



Assumptions and Constraints

1. Warehouse_ID uniquely identifies each WAREHOUSE and is assumed because the question provides no warehouse identifier.
2. Every WAREHOUSE contains exactly 200 STORAGE_UNIT entities, numbered with distinct integer values from 1 through 200.
3. Unit_Number is unique only within a particular WAREHOUSE, so STORAGE_UNIT is a weak entity.
4. Every STORAGE_UNIT belongs to exactly one WAREHOUSE and cannot exist independently of that WAREHOUSE.
5. The complete identifier of a storage unit is the composite key (Warehouse_ID, Unit_Number).
6. The shipping company has one or more warehouses; company-level attributes are outside the given requirements.
7. Complete key of STORAGE_UNIT = (Warehouse_ID, Unit_Number)